Data Availability and Source File Note

Companion to “Joint Signing-Partner and Audit-Firm Tenure and Environmental Disclosure: Evidence from Completing Partner Spells”

Prepared for submission to the Journal of Accounting Research

Deposited package: JAR_SourceData_CSVs

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Data Availability and Source File Note

1. Data Availability Statement (text for the Wiley Research Exchange submission form)

The findings of this study draw on third-party, commercially licensed data and on publicly available data; no human-subject or confidential personal data are used; the signing partners' names used to construct the tenure clock are public professional identifiers taken from published audit reports and are processed for no other purpose. Audit, financial, executive, and environmental-penalty records are from the CSMAR database, which is available only under a commercial license and cannot be redistributed by the authors; readers obtain these data through an institutional CSMAR subscription. The Huazheng ESG pillar scores used as the outcome and placebo measures, and the same environmental pillar as republished under a second reporting convention (E_W), are publicly available through Mendeley Data (Lin 2025, DOI: 10.17632/8fh8wtn6p5.2). Annual-report environmental-section text is taken from public exchange disclosures (Juchao Information Network, cninfo). In accordance with the Journal of Accounting Research Data and Code Sharing Policy, the authors provide a full description of every variable, the sample funnel, and the estimation scripts in Section 3 and Appendix A of the manuscript and in Sections 3 and 8 of this Note. The study-specific analysis code and the authors' constructed identifiers (the audit-spell ID and the signing-partner pair ID), together with the non-licensed derived data, will be deposited at the repository designated by the editors upon acceptance and given a persistent identifier to be cited in the article; the constructed data extract and code are available to the editor and reviewers on request during the evaluation period, subject to CSMAR licensing.

2. Summary of data sources

Data element	Source	Access	Role in the paper
Audit firm, signing CPA names, audit opinion	CSMAR audit database (1997-2025)	Commercial license (not redistributable)	Signing-pair construction, tenure clocks, rotation flags
Firm financial controls (size, ROA, growth, cash, leverage)	CSMAR fundamentals panel	Commercial license	Controls in all designs
Executive-structure covariates (gender, education, pay)	CSMAR executive panel	Commercial license	Controls (estimates insensitive to them)
Environmental penalty records	CSMAR environmental-penalty panel; firm-year wide file constructed by the authors from the licensed panel	Commercial license	Hard-outcome test (Section 4.7)
Huazheng E / S / G pillar scores	Huazheng ESG rating file	Public via Mendeley Data (Lin 2025)	Outcome (E) and placebo pillars (S, G)
Second E score (E_W)	Same Huazheng pillar under a second reporting convention	Public via Mendeley Data (Lin 2025)	Same-vendor, second-convention check only; not an independent rater
Annual-report environmental-section text	Public exchange PDFs (Juchao / cninfo)	Public	Textual test (Section 4.8)
State-ownership classification	CSMAR firm attribute	Commercial license	H4 heterogeneity

3. Treatment of third-party licensed data

The CSMAR audit, fundamentals, executive, and firm-attribute panels, and the penalty records derived from them, are licensed commercially and are therefore not redistributed by the authors, in line with the vendor license and the journal's policy on licensed data. To make the study verifiable without the licensed raw files, the deposited materials provide (i) the exact construction steps and code that turn a subscriber's CSMAR extract into every analysis variable, (ii) the variable definitions and the sample funnel (manuscript Section 3.1 and Appendix A), and (iii) all constructed identifiers. A reader with CSMAR access can therefore reproduce the estimation panel from the public Huazheng scores plus the licensed extract. The public components (Huazheng pillar scores and E_W via Mendeley Data; annual-report PDFs via cninfo) are independently downloadable and are cited with persistent identifiers.

4. Replication materials and the journal's Data and Code Sharing Policy

The authors follow the Journal of Accounting Research Data and Code Sharing Policy and its FAQ, administered through the journal's Chookaszian Accounting Research Center. Accordingly:

1. Full disclosure at submission. Manuscript Section 3 and Appendix A state every data source, the name-matching and clock-construction rules, the rotation-classification algorithm, the estimation specification (firm-and-year and pair-by-firm fixed effects, Frisch-Waugh-Lovell within-transformation, firm- and pair-firm-clustered inference), and the null-restricted Rademacher wild-cluster bootstrap (9,999 replications), sufficient for a reviewer to assess the design.

2. Deposit upon acceptance. The study-specific analysis code and the authors' constructed identifiers (the Symbol x cumulative-auditor-change spell ID and the two-name signing-partner PairID, which are the authors' work and not licensed CSMAR fields), plus the non-licensed derived data needed to reproduce every table, will be deposited at the repository designated by the editors upon acceptance and given a persistent identifier.

3. Availability during review. To allow verification of the reported estimates, the authors will make the constructed data extract and code available to the editor and reviewers on request during the evaluation period, subject to CSMAR licensing.

5. Classification and estimands note

The mandatory/voluntary rotation classification is built on the individual partner clock (the cap is enforced on the individual signing partner, not on the pair as a whole): a change is mandatory when the longer-serving outgoing partner has completed five consecutive service years on the client, not when the pair as a whole has served five years. The tenure-path estimates use CPATenure, the longer-serving partner's consecutive-service count, reconstructed independently from the audit-record name sequence. CPATenure coincides with the reconstructed within-firm individual clock in every firm-year where the reconstruction can be checked: perfect agreement over the 3,471 audit-firm-change years, where both clocks restart by construction, and over all 28,736 of the 28,737 non-change firm-years with a signing-name record. The one unchecked firm-year (2018, no name record) is immaterial: dropping it leaves the deadline fall at 0.646 (t = 7.44; Section 8.1). Because the cap is regulator-enforced rather than reconstructed, the agreement figure is a data-construction equivalence (a four-name ceiling on agreement), not a direct cap measurement. Article 7 of the 2003 decision (the one-year deferral when both signing partners reach the cap together) concerns the two signing partners, not the audit-firm clock; the joint five-year point is an empirical coincidence of the two clocks, not a statutory construct, and the manuscript states this in Sections 2.1 and 2.2.

The paper reports two estimands and keeps them separate. The **reporting estimand** is the year-four-to-year-five deadline fall (the contrast) in the firm-and-year fixed-effects specification (manuscript Section 3.2): the raw partner-clock contrast, reported as a descriptive upper bound. The **title estimand** is the match-adjusted joint high-tenure contrast in the pair-by-firm specification, defined as in manuscript Section 3.2: the interaction-implied joint-cluster differential at the five-year point (+0.653, t = 5.66; manuscript Section 4.3), with its post hoc four-or-five-window widening (+0.622, t = 6.13) reported as descriptive and not to be confused with the pooled no-interaction estimate of +0.295 reported below. It is a difference between joint pair-firm clusters and other pair-firm clusters, not an estimate for the firm-year cell in which both clocks read five. That cell-level destination estimate is +1.192 (manuscript Table 6) and the joint-subsample estimate is +1.201 (manuscript Table 8); neither is the +0.653 title estimand, and the three differ because they are estimated on different units. It is estimated as the interaction-implied joint effect — the main effect of the two-way-demeaned year-four-minus-year-five contrast (D2) plus its interaction with a joint-cluster indicator — which keeps a common fixed-effect structure across all 7,751 pair-firm clusters; the joint-subsample estimate, which re-absorbs fixed effects within joint clusters only, is reported alongside as a robustness number, and the difference between the two is explained in manuscript Section 4.3. The single year-five level coefficient is reported as descriptive only, because it is a level measured against a year-one base and is sensitive to that base. The pooled pair-by-firm estimate is reported as the match-adjusted average, with the note that it is weighted by within-cluster identifying variation rather than by cells. No family-wise error-rate adjustment is imposed across H1-H5, which are nested statements about the same path rather than independent hypotheses; the exploratory families named below carry their own Bonferroni thresholds and those thresholds are applied only within each family. The exploratory families are adjusted exactly as manuscript Section 5 states: $|t| = 2.50$ over the four firm-clock thresholds of Appendix A.12, and $|t| = 2.69$ over the seven Table 9 heterogeneity cells.

6. Sample reconciliation (for reviewer cross-check)

The paper uses several distinct panels — the event-design sample, the clock-keyed sample, the primary firm-year sample, the pair-by-firm samples, and the penalty, textual, composition, and pillar samples — and they are not directly comparable because each design has a different completeness requirement.

- **Event-design sample (Tables 2-3):** 32,017 firm-years on 4,571 firms (2013-2023).
- **Clock-keyed sample (descriptive cells; spell/successor/individual-clock designs):** 32,208 firm-years on 4,666 firms (retains 191 firm-years lacking the rotation classification or the ST/financial flag but with a complete clock and controls).
- **Primary firm-year sample (Tables 4-6, 10, 12; firm-and-year specifications):** 31,274 firm-years on 4,166 firm clusters under the tenures-one-to-five restriction (31,734 rows on 4,626 firms before singleton absorption).
- **Pair-by-firm samples (Tables 8-9; pair-firm specifications):** 19,376 pair-firm cells in 7,751 pair-firm clusters; of these, the 970 clusters containing at least one joint five-year cell hold 2,622 cells, and the 6,781 without hold 16,754 cells.
- **Penalty samples (Section 4.7):** 19,391 / 16,317 / 16,398 firm-years by forward window after listwise deletion and singleton absorption.
- **Textual samples (Section 4.8):** 762 sections matched to an E score; 779 sections from 365 firms in the firm-and-year contrasts.
- **Composition samples (Section 4.4, Table 10):** balanced panel of 13,122 firm-years on 1,203 firms observed in all eleven years (tenures one-to-five; 13,233 including year-six-plus rows); 271 pair-firm clusters with a joint cell belonging to balanced firms (897 cells).
- **Pillar samples (Section 4.5, Table 11):** 93,822 pillar-firm-year cells on the 31,274 firm-years, stacking E, S, and G.
- **Two-clock destination-cell sample (Section 4.3, Table 6):** the same 31,274 firm-years, with each firm assigned to the cell its audit-firm clock occupies in its first clock-five year: 1,764 joint firms, 463 with the firm clock below five, 996 with it above five, and 943 with no clock-five year inside the window.

- **Joint-placebo subsamples (Section 4.3, Table 7):** 7,452 / 6,376 / 6,247 / 5,594 / 515 firm-years on 1,716 / 1,431 / 1,885 / 2,066 / 235 firms for tenure years two to six respectively, each restricted to firms with both clocks at that tenure year.
- **Turnover-shock sample (Section 4.12):** 33,343 firm-years on 4,754 firms (2013-2023); partner exits reconstructed from 119,737 firm-partner-year records on 12,015 partners (2011-2023), giving 4,288 deadline-exit, 5,636 early-departure, and 9,822 reassignment firm-years, plus 3,674 audit-firm-change firm-years.
- **Matched-spell sample (Section 4.13):** 8,984 firm-years on 1,777 firms; all 1,611 treated spells match one-to-one with replacement (caliper 0.2 SD of the logit) to 839 unique control spells, controls drawn repeatedly (Section 8.22); 3,519 complete spells before matching (1,611 treated, 1,908 control) out of 7,456 spells on 4,666 firms.

Supporting counts. Tenure cells in the clock sample: 5,799 / 7,816 / 6,861 / 6,206 / 5,052 / 474 (year 6+), summing to 32,208. The mandatory-rotation flag marks 4,925 firm-years, of which 1,479 are full changes out of a clock-five year. Joint firm-year cells (both clocks at five) are 2,018 of the 31,274 primary firm-years. Signing-pair identities: 15,719 distinct pairs in the clock sample, against 15,011 individual signing partners counted over the full 1997-2025 audit panel. Pillar triple-difference: 93,822 cells on 31,274 firm-years.

7. Data statement

No human-subject or confidential personal data are used; the signing-partner names used to construct the tenure clock are public professional identifiers from published audit reports. All records concern listed-firm audit relationships and publicly or commercially licensed financial and ESG data.

8. Construction and reconciliation detail

8.1 Transition audit, full decomposition

Among the 24,696 within-spell transitions in the clock sample with an observable prior-year clock, 16,310 (66.0 percent) advance exactly one year, 297 (1.2 percent) skip over a missing panel year, 1,951 (7.9 percent) leave the clock unchanged, 5,995 (24.3 percent) move it backward and 143 (0.6 percent) jump forward by more than the calendar gap. The five cells are mutually exclusive and sum to 24,696, so the audit is an exact partition of its base. The non-advancing cells are dominated by pair replacements: 98.3 percent of them coincide with a change in the signing pair, in which the constructed clock follows the longer-serving partner and an incoming partner's unobserved prior service at the client can leave it above one or drop it below the outgoing count. The narrow partial-replacement margin — exactly one signing name changes while the other continues — accounts for 728 transitions, of which 235 keep the constructed clock consistent with the continuing partner's prior count plus one and 493 do not, 251 of the 493 spanning a calendar-year gap in the observable record. None of this residual moves the result: excluding the 493 non-conforming partial replacements leaves the deadline fall at 0.656 (t = 7.41; N = 30,781), and excluding all 728 partial replacements leaves it at 0.653 (t = 7.35; N = 30,544 - 730 rows leave the sample in total, the 728 flagged rows plus rows of firms that become singletons after the exclusion); both rows are in the manuscript's robustness table. Reconstructing each partner's name-matched history across the 2003-2025 name panel agrees with CPATenure in every one of the 32,207 firm-years where a name record exists; one firm-year (2018) carries no signing-name record and cannot be checked. Dropping that firm-year leaves the deadline fall at 0.646 (t = 7.44), unchanged to four decimals, so clock misclassification at this margin does not carry the result.

8.2 Lead-sample construction

The lead samples are smaller than the estimation sample because a lead is defined only where the deadline is observable within the panel (the firm must have a linkable future year): the one-year-ahead and two-year-ahead indicators are positive on 3,988 and 3,137 firm-years respectively, with 90.3 percent of one-year-ahead observations at clock four and 75.0 percent of two-year-ahead observations at clock three. The tests are informative at that margin: their minimum detectable effects are 0.424 and 0.354 points, both below the primary fall of 0.646, so the nulls bound early anticipation rather than merely being uninformative.

8.3 Event-study construction

Events are full pair replacements (both signing names replaced; 15,451 consecutive-year events in the clock sample) or full changes out of a clock-five year with the new pair's clock at one (1,479 events; event -1 is the deadline year for every event). Each firm-year is assigned to the earliest event whose [-4, +5] window contains it; the omitted category is event -1; the specification is firm and year fixed effects with controls, firm-clustered. Cell counts: 27,149 event cells across 4,035 firms (all changes) and 8,236 event cells across 1,392 firms (exact-five changes). The partner clock at event -1 is five for all exact-five events, and event -2 sits at clock four for 95.5 percent of its observations. The estimated cells are individually insignificant in the exact-five panel (event -2: +0.250, t = 0.74; minimum detectable effect 0.95 points); in the all-changes panel the only individually significant cell is the change year itself (-0.541, t = -3.56). Consistent with the cycle living at the cap, the fall is carried by firms that face it: among ever-mandatory firms the fall is 0.681 (t = 7.42; N = 25,267), while among never-mandatory firms it is 0.121 (t = 0.30; N = 6,007).

8.4 Penalty windows and N reconciliation

Penalty records exist on 25,749 firm-year rows in the extraction file, which covers 2013-2020. The current-year model needs a penalty record in the estimation year; the one-year-forward model drops the final panel year; the one-or-two-year-forward model flags a penalty in either forward year. Running all three columns on the extraction file reproduces the reported linear coefficients exactly (-0.0004, +0.0009, +0.0013). One provenance, one set of counts. The pre-dropna merge rows are 20,077 (current-year), 16,771 (one-year-forward), and 16,849 for the one-or-two-year-forward union, of which 78 rows enter only through the two-year-forward merge and carry missing clock and control values, so listwise deletion leaves 16,771; after singleton absorption the final estimation samples are 19,391 / 16,317 / 16,398 firm-years with 203 / 189 / 311 flagged cells. The 20,077 is the register merge on the clock-keyed sample: of the 20,418 clock-sample firm-years in 2013-2020, 20,077 carry a register record, and the merge yields 20,077 pre-dropna rows. The 20,079 quoted in Section 8.23 is a different base: the primary sample's own 2013-2020 row count, which is the base for the announcement-based measure. The two differ by two firm-years because the primary-sample restrictions remove 332 of the 20,077 register rows while 334 primary firm-years carry no register record (19,745 rows carry both). The base rates computed on those final samples are 1.05 / 1.16 / 1.90 percent (203/19,391 = 1.05; 189/16,317 = 1.16; 311/16,398 = 1.90), and these are the figures reported in manuscript Section 4.7. The 261 events are distinct penalty firm-years; 203 sit in a current-year position. Because the extraction panel ends in 2020, the one-or-two-year-forward indicator is mechanically truncated for 2019-2020 observations.

Table 8.4. Register counts by base: every count states its sample. The 203 of the penalty models is the current-year count on the 19,391-row estimation sample; the 204 is the same register counted on the full 20,079-row primary-sample window (of which the estimation sample is a subset); the two are not in conflict.

Base (register counts)	Firm-years	Flagged / register events (rate)	Note
Extraction file rows, 2013-2020 (file 05)	25,749	261 with a penalty	Vendor extraction
Register merge on the clock-keyed sample	20,077	current-year merge rows	Pre-listwise
Primary-sample firm-years 2013-2020	20,079	204 register firm-years (1.02 percent)	Common window with the broad flag
Rows carrying both primary-sample and register records	19,745	overlap	Section 8.4
Current-year estimation sample	19,391	203 flagged (1.05 percent)	Listwise deletion + singleton absorption
One-year-forward estimation sample	16,317	189 flagged (1.16 percent)	Drops final panel year
One-or-two-year-forward estimation sample	16,398	311 flagged (1.90 percent)	Union window

8.5 Provenance and deposit

Audit data, financial controls, executive covariates, and penalty records come from CSMAR (commercially licensed). The Huazheng pillar scores are public via Mendeley Data (Lin 2025); E_W is the same Huazheng pillar under a second reporting convention (same vendor, not an independent rating). Annual-report PDFs are public exchange disclosures. Authors cannot redistribute licensed CSMAR data; upon acceptance, the study-specific analysis code, the authors' constructed identifiers, and the non-licensed derived data needed to reproduce every table will be deposited at the repository designated by the editors and given a persistent identifier, subject to vendor license terms.

8.6 Abandoned constructions, for reproducibility

(1) A rating-divergence greenwashing index, abandoned because E_W and E_H are the same measure up to rounding and revision noise (correlation 0.99999997 on the raw extract; identical in 67.3 percent of firm-years), so they are not two independent methodologies. (2) Environmental keyword hit counts, dominated by template noise; they load on report length. (3) A narrow penalty definition, only 71 events, too few for powered inference. (4) A carbon-exposure measure, available for only 301 observations, replaced by a pollution-intensive industry indicator.

8.7 Sample reconciliation by design

Table 8.7 lists every estimation sample with its unit. Where a design is described in the manuscript, the manuscript table is named; the remaining rows are descriptive or construction samples.

Design	**N**	**Firms / cells**
Event estimation sample (Table 3)	32,017	4,571 firms
Clock-keyed sample (descriptive cells; spell/successor designs)	32,208	4,666 firms
Primary firm-and-year model, tenure 1-5, post-absorption (Tables 4-6, 10, 12)	31,274	4,166 firm clusters; 31,734 rows / 4,626 firms pre-absorption
Industry-by-year model (Table 12)	31,218	4,143 firm clusters (56 firm-years lack industry codes)
Pillar triple-difference (Section 4.5, Table 11)	93,822	pillar-cells on 31,274 firm-years
Two-way pair-and-firm model (Table 12)	15,800	firm-years with pair variation
Pair-by-firm model, pooled (Table 8)	**19,376**	**7,751 pair-firm clusters**
Pair-by-firm model, joint five-year clusters (Table 8)	**2,622**	**970 pair-firm clusters**
Pair-by-firm model, all other clusters (Table 8)	16,754	6,781 pair-firm clusters
Balanced panel (Section 4.4, Table 10)	13,122	1,203 firms observed in all eleven years, tenures 1-5
Event study, all changes (Table 13, Panel A)	27,149	event cells, 4,035 firms
Event study, exact-five changes (Table 13, Panel B)	8,236	event cells, 1,392 firms
Penalty LPMs (Section 4.7)	19,391 / 16,317 / 16,398	by forward window (Section 8.4)
Textual contrasts (Section 4.8)	779	365 firms; 762 sections matched to E
Mandatory-flag firm-years (clock sample)	4,925	exact-five full changes: 1,479
Turnover-shock sample (Section 4.12, Appendix A.10)	33,343	4,754 firms; 12,015 partners (2011-2023)
Matched-spell sample (Section 4.13, Appendix A.11)	8,984	1,777 firms; 1,611 treated spells matched to 839 unique controls (1:1 PSM with replacement)

8.8 Spell and pair construction

Contract-spell fixed effects use Symbol x a cumulative auditor-change counter; partner-pair fixed effects use PairID, the two-name signature as recorded. In the 32,208-row clock sample there are 15,719 distinct signing pairs; 7,314 serve more than one firm-year, of which 3,585 serve more than one firm (15,800 observations; the pair-FE subset). The 15,011 figure counts individual signing partners over the full 1997-2025 audit panel, a different object and a different population from the clock sample (which contains 9,628 distinct names). Partner-history reconstruction across the 2003-2025 name panel agrees with CPATenure in every firm-year where a name record exists; the single unchecked firm-year and the exclusion test are in Section 8.1 (the audit-firm-change-year split is quantified in Section 5 of this Note).

8.9 Rotation-classification boundary

A change is mandatory when at least one outgoing signing partner has completed five consecutive service years; the flag marks 4,925 firm-years. Of these, 1,479 are full changes out of a clock-five year (both names replaced, the new pair's clock at one, all flagged); the remainder are partial replacements in which a cap-reached partner departs while the shorter partner continues, plus 104 changes (2.1 percent) whose prior pair clock is below five and which are consistent with unobserved pre-2013 or cross-pair service under Article 4. Boundary cases such as the Article 7 deferral appear as year-six-plus observations and are excluded from inference.

8.10 Text-sample sizes

Four nested sizes underlie the textual results: 1,119 PDFs retrieved; 969 usable sections; 931 in the descriptive cells (653 + 175 + 103; the remaining 38 have an unclassifiable clock); 762 matched to a Huazheng E score. The firm-and-year contrasts use the 779 sections contributed by the 365 firms with more than one section (596 sections from 282 firms if the sample is instead restricted to the 762 matched sections).

8.11 Textual-evidence construction

If the year-five fall is a disclosure phenomenon, the annual-report environmental section should shrink in the final pre-deadline year. The PDF subsample is limited by source availability: of 1,119 full-text PDFs, about four-fifths date from 2013-2014 and none from 2021-2023; heading detection with a keyword fallback recovered 969 usable sections, covering 3.0 percent of estimation firm-years and over-representing early years. It validates the outcome's construction: across the 762 sections matched to a Huazheng E score within the estimation sample, E correlates with section length ($r = 0.24$), quantitative density ($r = 0.09$), and negatively with commitment-word density ($r = -0.22$); the boilerplate ratio is essentially orthogonal ($r = 0.01$). In firm-and-year fixed effects on those 779 sections, with outcomes standardised and standard errors clustered by firm, no clock-five contrast is distinguishable from zero: year five minus year four is -0.13 ($t = -1.09$) for section length, -0.02 ($t = -0.13$) for quantitative density, +0.20 ($t = +1.93$) for commitment-word density and -0.18 ($t = -0.91$) for the boilerplate ratio, with minimum detectable effects of 0.29 to 0.55 standard deviations, 3.2 to 6.1 times the standardised rating fall of 0.09. The paired clock-four-to-clock-five commitment contrast across the 40 firms seen in both years is +0.08 per 1,000 characters ($t = 0.13$), so the exploratory margin that the unpaired year-five/other comparison suggests ($z = 1.11$ on all 969 sections) does not survive a within-firm contrast.

8.12 Year-by-year deadline-fall cells

The primary policy-descriptive number is Section 8.17's pooled repeal interaction: +0.816 (t = 7.84) before the repeal against +0.069 (t = 0.36) after, a difference of +0.747 (t = 3.26). The 2022-only and 2023-only cells behind the 2022-2023 window are year-interacted tenure-dummy contrasts estimated on the pooled window (firm and year FE, controls, firm clusters; tenures one-to-five, N = 7,134 after singleton absorption): 2022 = -0.44 (t = -1.67), 2023 = +0.38 (t = 1.19). Both cells remain indistinguishable from zero. Under the alternative fiscal-year coding the post window is 2023 alone, the same cell. Sliding two-year windows (Section 4.9 rows, tenures one-to-five) run from -0.063 (2013-2014) to +1.374 (2020-2021), peak at 0.934 (2017-2018), and turn null once a window touches the repeal (2021-2022 -0.077, 2022-2023 -0.059). Three-year windows are reported in the deposited replication output: the three-year fall is significant in every window from 2014-2016 through 2020-2022 and null in the window that straddles the repeal (2021-2023: -0.080, t -0.95).

8.13 Restricted versus unrestricted wild-cluster bootstrap on the pair-by-firm estimates (9,999 draws)

The pooled pair-by-firm fall of 0.295 (cluster-robust t = 2.27, CRV1, 19,376 cells in 7,751 pair-firm clusters) was resampled with Rademacher weights under both wild-bootstrap variants. The restricted (WCR) variant draws residuals from the model estimated without the deadline contrast, so the resampling is centered on the null; the unrestricted (WCU) variant draws residuals from the full model and is centered on the estimated alternative. With clusters this small (average 2.5 cells; 5,038 of 7,751 clusters have exactly two cells), the WCU distribution inherits the estimated effect rather than the null: 98.6 percent of its bootstrap statistics are positive and its 2.5th percentile (0.26) already exceeds zero. A test centered away from zero cannot be read as a p-value against the null, which is why the unrestricted p of 0.51 does not contradict the cluster-robust t of 2.27. The restricted variant is the appropriate null test and returns a two-sided p of 0.029 (one-sided 0.015) at 9,999 draws, with a bootstrap distribution centered at zero (50.3 percent positive draws). The same restricted procedure applied to the title estimand returns: joint five-year clusters 1.201 (t = 4.15; WCR $p < 0.001$), all other clusters -0.161 (t = -1.09; WCR p = 0.27), and the interaction specification D2 = +0.057 (t = 1.02; WCR p = 0.30), D2 x joint = +0.596 (t = 4.63; WCR $p < 0.001$), implied joint effect D2 + D2 x joint = +0.653 (t = 5.66; WCR $p < 0.001$).

8.14 Pair-by-firm diagnostics: year-three placebo and cluster-level checks

Placebo. Replacing the year-four-to-year-five contrast with the year-two-to-year-three contrast in the same pair-by-firm specification gives -0.163 (t = -1.46, N = 19,376), consistent with no pre-deadline step at the pair-firm margin. Cluster influence, single definition. For each cluster define the score c_g as the sum over its cells of the two-way-demeaned year-four-minus-year-five indicator times the pair-by-firm residual. The signed cluster scores sum to exactly zero by construction (the residual is orthogonal to the demeaned regressor), so shares of that net total are undefined; measured against the total absolute score mass (the sum of absolute cluster scores over the 7,751 clusters), the five largest clusters hold 1.0 percent, the ten largest 1.8 percent, and the top decile (776 clusters) 51.4 percent; 47.4 percent of clusters carry positive scores, so the result is not an artifact of a handful of one-sided clusters. Excluding the five largest-score clusters (10 cells; N = 19,366 across 7,746 clusters) leaves the fall at 0.276 (t = 2.17); excluding the ten largest (23 cells; N = 19,353 across 7,741 clusters) leaves it at 0.266 (t = 2.11), and the restricted wild-cluster bootstrap on that excluded sample gives a two-sided p of 0.032 (9,999 replications). Collapsing to one observation per cluster, the cluster-level WLS contrast is 0.90 (t = 2.35) and the unweighted OLS contrast 0.70 (t = 1.86) across 7,751 clusters. Cluster sizes: 5,038 clusters with 2 cells, 1,812 with 3, 686 with 4, 183 with 5, 23 with 6, 5 with 7, 4 with 8 (no singletons by construction).

8.15 Incoming-partner prior service (name matching)

The data carry no CPA license identifiers, so prior service must be recovered by name matching. Substituting name matching across the 3,471 audit-firm-change years: 1,282 firm-change years (36.9 percent) are flagged when any incoming partner's name appears in an earlier year at the same client, and 582 when all incoming partners' names do. The distribution of the maximum prior same-client service among the 1,282 flagged years is: 1 year, 242; 2, 205; 3, 230; 4, 204; 5, 170; 6, 105; 7, 48; 8, 36; 9, 28; 10, 10; 11, 3; 12, 1 (median 3, interquartile range 2 to 5). Excluding the 1,282 flagged years leaves the fall at 0.726 (t = 8.11; the flagged count sits on the clock-keyed sample, and 1,244 rows leave the 31,274-row primary sample, N = 30,030); excluding their spells leaves 0.784 (t = 8.09); excluding every firm-change spell leaves 0.823 (t = 7.69).

8.16 Within-pair-firm heterogeneity: joint five-year clusters and score concentration

Cluster split, and which estimate carries the title claim. The pair-by-firm fall of 0.295 (t = 2.27; 19,376 pair-firm cells in 7,751 clusters; cluster-robust SE 0.130) was decomposed along the joint-clock dimension. The title estimand is the interaction-implied joint effect: entering the deadline fall contrast (D2, the two-way-demeaned year-four-minus-year-five contrast that carries the fall) and its interaction with a joint-cluster indicator, with pair-firm and year fixed effects, controls, and pair-firm clustering over all 7,751 clusters, returns D2 = +0.057 (t = 1.02; WCR p = 0.30), D2 x joint = +0.596 (t = 4.63; WCR p < 0.001), and an implied joint effect of D2 + D2 x joint = **+0.653 (t = 5.66; restricted WCR p < 0.001; MDE 0.32)**. This is the headline number because it keeps a common fixed-effect structure across all clusters. Re-estimated separately over the pair-firm clusters that contain at least one joint five-year cell (both clocks at five within the same cluster) and over all others, the fall is 1.201 (t = 4.15; restricted WCR p < 0.001; MDE 0.81) in the 970 joint clusters (2,622 pair-firm cells) and -0.161 (t = -1.09; restricted WCR p = 0.27; MDE 0.41) in the remaining 6,781 clusters (16,754 cells). The subsample estimate is the larger of the two because estimating on the joint subsample alone re-absorbs the pair-firm and year fixed effects within joint clusters only, which reweights the identifying variation toward within-joint-cluster contrasts; both specifications agree that the fall lives in joint clusters and nowhere else.

The pooled 0.295 is not the cell-weighted average of the two subsample estimates (0.135 x 1.201 + 0.865 x (-0.161) = 0.02): the pair-firm and year fixed effects are re-absorbed within each subsample, so the pooled estimate does not decompose into a cell-share-weighted combination; the joint clusters contribute 28.0 percent of the deadline-indicator variation (the sum of squared two-way-demeaned year-four-minus-year-five indicators) on 13.5 percent of the cells. The 2,622 cells here are pair-by-firm cells of the 19,376-cell pair-by-firm estimation sample; the 2,018 count refers to firm-year observations in the joint cell of the 31,274-row firm-year sample - a different unit.

Power. The pair-by-firm design has MDE = 2.8 x SE = 2.8 x 0.130 = 0.363 points, so the design would detect a fall of the joint-cluster magnitude with ease but is powered to about a third of the primary estimate; the null in non-joint clusters bounds their fall below 0.36 points. The title estimand's MDE is 0.32 points; the joint subsample MDE is 0.81 points and the non-joint subsample MDE 0.41 points.

Score concentration (same single definition as Section 8.14: c_g = the sum over the cluster's cells of the two-way-demeaned year-four-minus-year-five indicator times the pair-by-firm residual, cluster = pair-firm). The distribution across the 7,751 clusters is close to symmetric: deciles 10/25/50/75/90 at -2.863 / -0.577 / -0.005 / 0.574 / 2.962, SD 3.65, with 47.4 percent of clusters carrying positive scores (holding 50 percent of absolute mass by the symmetry just noted). Concentration is limited under the absolute-mass definition: the five largest-magnitude clusters account for 1.0 percent and the ten largest for 1.8 percent of total absolute score mass, so the pair-by-firm estimate is not an artifact of a handful of clusters; its heterogeneity runs along the joint-clock dimension instead - the 970 joint clusters hold 28.8 percent of absolute score mass and 28.0 percent of the deadline-indicator variation, on 13.5 percent of the cells.

8.17 Composition: the balanced panel and the spell-completion interaction (manuscript Section 4.4, Table 10)

Balanced panel. The balanced panel is the set of firms with eleven observed years in the 2013-2023 clock sample: 1,203 firms. On the tenures-one-to-five window it holds 13,122 firm-years and the year-four-to-year-five contrast is -0.113 (t = -0.89; SE 0.127; MDE 0.355 at 80 percent power), with year coefficients -0.024 / -0.030 / +0.048 / +0.161 (t -0.19 / -0.20 / 0.29 / 0.89). The panel is not starved of cap observations: 2,622 year-four cells (a balanced-panel row count, unrelated to the 2,622 pair-firm cells of Section 8.16), 1,887 year-five cells, 413 joint cells (both clocks at five), and 86.9 percent of balanced firms (1,046 of 1,203) reach clock five at least once. Restricting to balanced firms that reach clock five leaves the fall at -0.098 (t = -0.77; 1,038 firms, 11,321 rows; MDE 0.357).

Interaction specification. Adding the year-two-to-year-five indicators interacted with a balanced-firm indicator to the primary-sample regression (firm and year FE, controls, firm-clustered; N = 31,274 on 4,166 firms) returns Y2 x balanced = -0.165 (SE 0.163, t = -1.02), Y3 x balanced = -0.221 (SE 0.198, t = -1.11), Y4 x balanced = -0.788 (SE 0.220, t = -3.58) and Y5 x balanced = +0.522 (SE 0.230, t = 2.27). The implied levels against the year-one base are, for non-balanced firms, +0.140 / +0.181 / +0.839 / -0.359 at years two to five and, for balanced firms, -0.025 / -0.040 / +0.051 / +0.163. The implied fall is +1.198 (SE 0.118, t = 10.19; MDE 0.33) among non-balanced firms and -0.112 (SE 0.127, t = -0.88; MDE 0.356) among balanced firms, so both the year-four elevation and the year-five drop are individually and significantly reshaped for persistent firms. The two interactions that the deadline cannot reach (year two and year three) are jointly indistinguishable from zero (Wald chi-square(2) = 1.38, p = 0.50).

Pre-trend check by balanced status. The implied year-two-to-year-three contrast is -0.040 (SE 0.099, t = -0.41) among non-balanced firms and +0.015 (SE 0.114, t = 0.14) among balanced firms; the difference between the two contrasts is -0.056 (SE 0.151, t = -0.37), with a minimum detectable difference of 0.423 points at 80 percent power. The two groups therefore do not diverge before the deadline can bind. The divergence begins one step later: the year-three-to-year-four difference-in-differences contrast is -0.567 (SE 0.162, t = -3.51) and the year-four-to-year-five contrast is +1.310 (SE 0.173, t = 7.56). This ordering is what the deadline reading predicts and what a difference-in-levels reading does not, since the descriptive gaps between the two groups are level differences that would appear from year two onward. The check bounds the selection concern rather than eliminating it: it rules out a pre-divergence larger than 0.42 points but cannot say why persistent firms sit at a different level.

Composition correction. Matching the two groups on size, ownership, listing age and leverage (2,721 firms, 25,683 rows) leaves the non-balanced fall at +0.934 (t = 6.56) against -0.118 (t = -0.93), a gap of 1.052 (t = 5.52) against 1.310. Entropy balancing on ten firm means cuts the gap to 0.514 (t = 1.40) but concentrates the weights on listing age, near the group's definition; after weighting there are 928 effective comparison firms against 2,402 before it, and the gap's minimum detectable effect at 80 percent power is 2.80 x its standard error (0.514/1.40), or 1.03 points. That entry is now the ninth row of manuscript Table 10, so the ten-covariate result cited in the abstract is tabulated rather than resting here alone. The firm-level counterpart of the Section 8.26 cluster reweighting, with persistence and spell-completion constraints added, is Section 8.27.

Repeal interaction. Pooling the pre-repeal years (2013-2021) against the two post-repeal years (2022-2023) in one regression that interacts the year-four and year-five indicators with a post-repeal indicator (firm and year FE, controls, firm-clustered; 23,597 pre-repeal and 7,677 post-repeal rows) gives Y4 x post = -0.726 (SE 0.214, t = -3.39) and Y5 x post = +0.021 (SE 0.235, t = 0.09). The implied fall is +0.816 (SE 0.104, t = 7.84) before the repeal and +0.069 (SE 0.194, t = 0.36) after it; the difference is +0.747 (SE 0.230, t = 3.26), with a minimum detectable difference of 0.642 points. This is the same before-and-after comparison reported window by window in Section 4.9 (0.755 in 2013-2019, 1.374 in 2020-2021, -0.059 in 2022-2023); the pooled interaction version is the more precise of the two because it identifies both contrasts against the full sample's fixed-effect structure rather than against the post-window rows alone.

Repeal first stage. Independence Standard No. 1 (2024) keeps the five-year cap and treats the second signing partner as an engagement partner, so the 2022 announcement removed an instrument, not the obligation. The signing panel shows no release: of 829 clients at clock five in 2022, 464 (56.0 percent) kept at least one signing partner into 2023, 20 kept both, 11 reached six years, and none did in 2024-2025; the share of client-years above five falls from 2.2 percent in 2018-2021 to 1.3, 0.5, 0.2 and 0.1 percent across 2022-2025. A repeal difference-in-differences therefore has no first stage; the post-repeal coefficient is a replication test.

Related composition rows (primary sample unless noted): firms whose observed span touches both window edges (2013-2023), 1,792 firms and 18,601 rows, fall -0.045 (t = -0.42; MDE 0.30); firms with at least ten observed years, 1,740 firms and 18,442 rows, +0.021 (t = 0.20; MDE 0.31); firms ever reaching clock five, 3,273 firms and 25,650 rows, +0.641 (t = 7.28; MDE 0.25); firms with all five tenure levels observed, 1,867 firms and 17,666 rows, +0.517 (t = 4.64; MDE 0.31).

Joint level in the balanced panel. Of the 3,352 pair-firm clusters belonging to balanced firms, 271 contain a joint five-year cell (897 cells); the joint-cluster fall among them is -0.096 (t = -0.23; SE 0.418; MDE 1.172), so the balanced-panel joint subsample is marginally powered against the full title estimate of 0.653 and its null is suggestive rather than decisive. Joint cells are 53.4 percent of year-five rows among non-balanced firms and 21.9 percent among balanced firms; the balanced share is 413 of 1,887 year-five rows, and the non-balanced share uses the same year-five-row denominator on the non-balanced subset. Manuscript Section 1 and Table 6A, Panel B quote both shares.

The full two-by-two behind those shares, on the primary sample's partner-clock-five firm-years (recomputed from file 18): balanced firms hold 1,887 clock-five rows, of which 413 are joint (21.9 percent); non-balanced firms hold 3,004 clock-five rows, of which 1,605 are joint (53.4 percent); together 4,891 clock-five rows carry the 2,018 joint firm-years, so the two shares and the joint-firm-year total reconcile exactly.

Descriptive comparison, balanced against full primary sample (firm-level means): size 22.56 / 22.12; SOE share 36.9 / 29.4 percent; age index 2.50 / 1.81; firm-change years 1.20 / 0.82; top-ten-auditor share 68.2 / 72.8 percent; ever at clock five 86.9 / 70.6 percent. Industry mix is close to identical: manufacturing 70.7 / 69.4 percent of firms, mining and utilities 4.6 / 4.2 percent, with the same three largest industries in both groups (C39, C26, C38 for balanced firms; C39, C26, C35 for the full sample).

Figure 1 data (residualized cell means, pair-firm and year FE for the pair rows, firm and year FE for the balanced row). Joint clusters (970 clusters, 2,622 cells): years 1-5 = -0.202 / -0.398 / -0.269 / +0.912 / -0.612 (cells 136 / 223 / 310 / 915 / 1,038). All other clusters (6,781 clusters, 16,754 cells): -0.078 / -0.025 / -0.046 / +0.103 / +0.078 (cells 2,390 / 4,461 / 4,525 / 3,547 / 1,831). Balanced panel (1,203 firms, years 1-5 rows): -0.032 / -0.038 / -0.042 / +0.019 / +0.147 (cells 2,231 / 3,295 / 3,087 / 2,622 / 1,887). Plotted values are residualized cell means, not regression estimates; the manuscript's regression contrasts are in Tables 8 and 10. The cell-level standard errors behind the plotted whiskers are regenerated by the deposited code from the main panel rather than archived as numbers; the archive records the cell means and counts (workbook sheet 23).

8.18 The two clocks at the partner clock's five-year point (manuscript Section 4.3, Table 6)

Each firm is assigned to the cell its audit-firm clock occupies in that firm's first clock-five year (firms with several clock-five years use the first). The level is measured against the year-one base and the fall is the year-four-minus-year-five contrast within the cell; the regression is E on cell-specific year-four and year-five indicators with firm and year fixed effects, full controls and firm clusters (N = 31,274 on 4,166 firms). Manuscript Table 6A reports the full distribution of the audit-firm clock across these cells; both clocks are in file 18 (CPATenure, AuditorTenure).

Cell (partner clock at five)	Firms	Year-five level	t	Fall	t
Audit-firm clock also at five (joint)	1,764	-0.547	-4.08	+1.192	8.85
Audit-firm clock below five	463	-0.120	-0.56	+0.104	0.52
Audit-firm clock above five	996	+0.221	1.32	+0.191	1.32
No clock-five year inside the window	943	-	-	-	-

The difference between the joint cell and the firm-clock-below cell is +1.088 points. Pooling the two years of the high-tenure window (firm clock four or five) gives a fall of +1.114 (SE 0.126, t = 8.85) on 1,892 firms and a year-five level of -0.520 (t = -3.80), against -0.020 (t = -0.08) below four and +0.191 (t = 1.33) above. The cross-sectional ordering matches the year-five levels in manuscript Table 6. The 943 firms with no clock-five year inside the window contribute no year-five cell and are identified only through year-four observations. Manuscript Table 5, the interaction version of the same comparison, is one firm-and-year fixed-effects regression on the same 31,274 firm-years with the year-two and year-three path indicators, both clock-four and clock-five dummies for each clock, the ten controls and firm clusters; its joint year-five cell is estimated in the mutually exclusive four-cell version of that specification (partner-five-only, firm-five-only and both-five replacing the two year-five singles).

8.19 Joint placebo by tenure year (manuscript Section 4.3, Table 7)

For each tenure year t the "both clocks at t" firms are selected and the year-(t-1)-to-year-t contrast is estimated within that subsample (firm and year FE, controls, firm clusters), reported in the manuscript's sign convention so that a positive number is a decline.

Tenure year t	Contrast	Estimate	t	Firms	MDE (80%)
2	Year 1 to year 2	-0.001	-0.01	1,716	0.30
3	Year 2 to year 3	+0.081	0.70	1,431	0.32
4	Year 3 to year 4	-0.130	-1.04	1,885	0.35
5	Year 4 to year 5	+0.938	6.81	2,066	0.39
6	Year 5 to year 6	-0.043	-0.07	235	1.82

The year-six subsample is thin (235 firms) and its null is uninformative rather than confirmatory. Note that the year-five entry (+0.938) is the year-four-minus-year-five contrast within the both-clocks-at-five subsample, whereas the Table 6 joint fall (+1.192) is the same contrast within the destination-cell joint firms; the two use different subsamples and are not directly comparable, but both locate the movement at the five-year joint point and nowhere else.

8.20 Entropy balancing and exploratory mechanism splits (manuscript Section 4.3 and Appendices A.8-A.9)

Entropy balancing (Hainmueller and Xu 2013) reweights the 2,402 non-joint firms to match the 1,764 joint firms exactly on sample means of state ownership, size, return on assets, growth, cash, leverage, age index, a top-ten-auditor indicator and a manufacturing indicator (the firm-level mean of an indicator for a two-digit industry code beginning with C). The dual objective converged at -0.207 (largest balance gap below 1e-7) and the weights lie between 0.20 and 4.70; every balance gap is zero to three decimals. On the entropy-balanced sample the joint fall is +1.182 (SE 0.135, t = 8.79) against +1.192 (SE 0.135, t = 8.85) unweighted, the firm-clock-below fall +0.216 (SE 0.223, t = 0.97) against +0.104 (SE 0.199, t = 0.52), and the firm-clock-above fall +0.254 (SE 0.155, t = 1.64) against +0.191 (SE 0.144, t = 1.32); the joint-minus-firm-clock-below difference narrows from +1.088 to +0.972.

The three exploratory mechanism splits (all unadjusted for multiple testing, all using the Table 6 destination-cell specification with the joint-cell year-four and year-five indicators interacted with the moderator) are: top-ten auditor +0.859 (t 4.45) against +1.365 (t 4.48), difference -0.506 (t -1.57); industry specialist (auditor's client share in the firm's two-digit industry-year above the 0.0796 median) +0.898 (t 3.97) against +1.102 (t 4.75), difference -0.204 (t -0.71); top-quartile client importance (firm assets as a share of auditor portfolio assets above 0.0100) +0.416 (t 1.06, MDE about 1.1) against +1.111 (t 5.97), difference -0.695 (t -1.71). The fall is present in both arms of every split and is if anything smaller where audit-firm review machinery should bind more tightly; the splits bound the mechanism and do not support a firm-review-cycle reading.

8.21 Turnover-shock design (manuscript Section 4.12 and Appendix A.10)

Partner exits are reconstructed from the signing-partner name strings of the audit panel over 2011-2023: strings are split on the delimiters used in the file and de-duplicated within firm-year, giving 119,737 firm-partner-year records on 12,015 partners and 5,623 firms. Individual tenure counts consecutive years in which a name signs the same client, restarting after a gap. An exit is the last year of a run, retained only when the client is observed in the following year and the year is not the panel's last, giving 36,448 exits. Exits at individual tenure five or more are deadline exits (5,943 partner-client exits, 4,288 firm-years after collapsing); the remainder are early exits, split into reassignments when the name signs another client the next year (19,962 exits, 9,822 firm-years) and departures when it does not (10,543 exits, 5,636 firm-years). Audit-firm changes come from the panel's change indicator (3,674 firm-years). Leads and lags are formed within firm with groupwise shifts; the base category is all firm-years outside each shock's two-lead/two-lag window.

Cross-checks for a replicator: (i) the 36,448 exits exceed the firm-year counts because two partners sign most clients and both may exit in the same year; (ii) a first version required a clean window with no other exit within three years, which retains only 180 departures on 57 firms with uninformative cells (minimum detectable effects 2.4 to 6.7 points), so the deposited construction is the unrestricted within-firm specification; (iii) 92.1 percent of deadline-exit firm-years sit at team clock five, which is why the deadline indicator is reported as a proxy for the clock and not as an independent instrument; (iv) the one-year-ahead lead on the deadline shock is mechanically the year-four peak and is not a pre-trend: 83.5% of the firm-years one year before a deadline exit sit at team clock four, which is why the lead recovers the year-four coefficient rather than measuring anticipation.

8.22 Propensity-score matching of tenure spells (manuscript Section 4.13 and Appendix A.11)

Spells come from the spell field of the estimation file (7,456 spells on 4,666 firms); a spell is complete if the firm is observed in a later year under a different spell or the spell ends before 2023, leaving 3,519. Covariates are the ten controls and the E score, taken at the spell's first observed year (the tenure-minimum row within the spell), plus opening-year dummies in the propensity model. The logit is estimated by iteratively reweighted least squares; matching is nearest-neighbour one-to-one with replacement inside a caliper of 0.2 standard deviations of the logit score, with the treated spells visited in a fixed random order (seed 20261021). All 1,611 treated spells match, drawing on 839 unique control spells; controls are repeated in the matched sample according to how many times they are drawn, which is the standard treatment of replacement. The largest absolute standardized mean difference falls from 0.712 (on the propensity score itself) to 0.045, and no covariate exceeds 0.10 after matching. Inverse-probability weights are p/(1-p) on control spells and one on treated spells. Replicators using a different caliper or a Mahalanobis metric should expect the fall to move within the 0.65 to 0.71 range reported in Appendix A.11; it should not be expected to reproduce to three decimals.

8.23 Why the penalty panel is not extended past 2020, and what a broader measure does

The annual-report extraction that produces the penalty file covers 2013-2020 only. The merged panel's environmental-penalty flag is populated for 2013-2020 and is recorded as zero for every 2021-2023 row, which is absence of extraction rather than absence of penalties, and the same is true of the disclosure-hit count, whose 2021-2023 rows are entirely missing. Neither can therefore be extended from the deposited materials; a replicator would need the CSMAR environmental-penalty table or an equivalent enforcement register for 2021-2023. Extending the panel would in any case not resolve the power problem: 2021-2023 adds 11,195 firm-years to the 20,079 firm-years the primary sample holds in 2013-2020 (Section 8.4 reconciles this base with the 20,077-row register merge), a 1.56-fold enlargement that shrinks standard errors to 0.80 of their present size and moves the narrow-definition minimum detectable effect from 0.66 to 0.53 percentage points.

The merged panel's flag is broader than the penalty file's count — 1,414 events on the primary sample's 20,079 firm-years in 2013-2020 (7.04 percent) against the register's 204 firm-years on the same window (1.02 percent) — and its year-four-to-year-five contrast is +1.458 percentage points (SE 0.545, t = +2.68). That estimate is not used in the paper, because it is produced by two years: 73.7 percent of the events fall in 2017 and 2018, whose base rates are 17.6 and 19.9 percent against 1.5 to 5.6 percent elsewhere; excluding them gives -0.462 (t = -0.98), estimating on them alone gives +3.361 (t = +2.15), and leave-one-year-out estimates run from +1.969 (t = +3.18, dropping 2015) to +0.604 (t = +1.05, dropping 2017). The file also confirms that the narrow flag is a strict subset of the broad one (no firm-year flagged narrowly is unflagged broadly, over 21,006 overlapping rows), and that the broad flag and the disclosure-hit count agree as indicators in all 21,006 rows.

8.24 Cross-rater replication: not possible from the deposited materials

The archived E_W column is the same Huazheng rating under a second reporting convention, not a second rater: the archived copy equals E_H exactly in all 10,352 overlapping rows; in the raw cross-validation extract (deposited separately) 67.3 percent of firm-years are identical and the remainder differ by rounding or revision noise, the maximum absolute difference being 0.109 score points, so the correlation is 1.0000 to six decimals. The panel's rating-divergence column is the absolute difference between the two conventions (its ratio to |E_H - E_W| has median exactly 1.000), so it measures rounding and revision noise within one vendor, not disagreement between vendors. A cross-rater replication would require an independently constructed rating — SynTao, Rankins (RKS), Bloomberg, or FTSE — each of which is licensed separately and none of which is in the deposited package; the paper does not claim to have performed one and the single-rater limitation is stated in Section 5.

8.25 An independent, publicly reproducible penalty measure (2013-2023)

Table 8.25. Penalty measures: windows, denominators, events, and base rates. Each base rate states its denominator; the register and announcement measures are not pooled with one another (Section 8.25). Minimum detectable effects: register 0.61 to 0.72 percentage points on the estimation samples (0.66 on the 2013-2020 current-year window); announcement measure 0.486 on the 2013-2020 overlap and 0.370 on the full window.

Measure	Window; denominator	Events (base rate)	Source
Licensed register (file 05)	2013-2020; estimation samples 19,391 / 16,317 / 16,398	203 / 189 / 311 (1.05 / 1.16 / 1.90 percent)	CSMAR, licensed; Table 8.4
Register, common window	2013-2020; primary sample, 20,079 firm-years	204 firm-years (1.02 percent)	Section 8.4, Table 8.4
Broad enforcement flag	2013-2020; primary sample, 20,079 firm-years	1,414 firm-years (7.04 percent)	Section 8.23; not used as a finding
Announcement indicator (file 16)	2013-2023; 150 firm-years, 124 on the 31,274-row primary sample	124 firm-years on the primary sample (0.40 percent), from 203 announcements and 119 firms; 150 firm-years on the full 2013-2023 window	cninfo, public
Register-announcement overlap	2013-2020 overlap window	19 firm-years (correlation 0.134)	Section 8.25

The licensed penalty register in the deposited package stops in 2020, so Section 4.7 also reports an outcome that any replicator can rebuild without a licence. We query the announcement index of China's designated disclosure platform for listed firms (巨潮资讯网, cninfo; endpoint http://www.cninfo.com.cn/new/hisAnnouncement/query, posted year by year) for 2013-2023 with the search terms 处罚, 环保处罚, 环境处罚 and 行政处罚, de-duplicate on the announcement identifier (6,641 announcements from 306 requests), and keep announcements whose title - after deleting the firm's own short name - contains both a penalty instrument and an environmental authority or harm, while excluding titles naming securities, tax, customs, market-supervision, work-safety, fire or transport regulators. This yields 203 announcements from 119 firms and 150 firm-years; 124 of those firm-years fall on the 31,274-firm-year primary sample (a 0.40 percent firm-year rate). Files 16 and 17 deposit the firm-year indicator and every estimate behind this section, including the leave-one-year-out range. Two caveats travel with the measure. It is a disclosure-based outcome: disclosure duties bind on material penalties at the listed parent, so subsidiary-level decisions recorded in the register need not appear, and on the 2013-2020 overlap the two measures coincide on 19 firm-years (correlation 0.134). It is also sparse - a 0.396 percent base rate (rounded to 0.40 where the manuscript quotes it) - so it moves its own bound from 0.486 percentage points on the 2013-2020 overlap to 0.370 on the full window, against 0.66 for the register's own minimum detectable effect on the same 2013-2020 overlap (Section 4.7, Appendix A.4). It is a robustness check on the penalty nulls, not a replacement for the register; the two are not pooled.

8.26 Placebo joint thresholds and cluster-level reweighting (manuscript Sections 4.3 and 4.10, Appendix A.12)

Firm-clock placebo (Table A.6). The joint cell is re-cut on the audit-firm clock the firm carries in its first partner-clock-five year, and the year-four-to-year-five fall is re-estimated in each cell with the Table 6 partition specification (one firm-and-year fixed-effects regression on the 31,274 primary firm-years, controls, firm-clustered). Re-estimated with the Table 6 specification, the benchmark reproduces the joint cell exactly (+1.192, t = 8.85); +0.104 (t = 0.52) below five and +0.191 (t = 1.32) above. Cells: firm clock 4 +0.423 (SE 0.324, t 1.31; 128 firms; MDE 0.91); 5 +1.192 (SE 0.135, t 8.85; 1,764 firms; MDE 0.38); 6 +0.215 (SE 0.357, t 0.60; 173 firms; MDE 1.00); 7 +0.228 (SE 0.304, t 0.75; 157 firms; MDE 0.85); all other values +0.170 (SE 0.145, t 1.17; 1,944 firms; MDE 0.41). Joint minus placebo: 4 +0.769 (t 2.19), 6 +0.976 (t 2.55), 7 +0.964 (t 2.90), all other values +1.022 (t 5.16). The firm-clock distribution at the clock-five year runs from 1 (101 firms) to 19 (1 firm); cells with fewer than 40 firms are not estimated.

Pair-by-firm placebo. The coincidence is redefined at firm clock tau as a cluster-level indicator, and Table 8 is re-estimated (pair-firm and year fixed effects, pair-firm-clustered; D2 is the two-way-demeaned year-four-minus-year-five contrast). D2 x joint / implied effect: tau = 4, +0.208 (t 1.33) / +0.417 (SE 0.148, t 2.82; 137 clusters); tau = 5, +0.596 (t 4.63) / +0.653 (SE 0.115, t 5.66; 970 clusters); tau = 6, -0.169 (t -0.63) / +0.056 (SE 0.263, t 0.21; 157 clusters); tau = 7, -0.455 (t -2.47) / -0.216 (SE 0.177, t -1.22; 159 clusters). Restricted to clusters holding no joint five-year cell: tau = 4, +0.427 (t 2.59) / +0.457 (SE 0.155, t 2.96; 135 clusters); tau = 6, -0.011 (t -0.04) / +0.045 (t 0.17); tau = 7, -0.290 (t -1.59) / -0.215 (t -1.24). Overlap with the joint-five clusters: 2 of 137 at tau = 4 (1.5 percent), 0 at tau = 6 and 7. Robustness of the tau = 4 estimate: cell-level flag +0.845 (t 2.18); clusters with at least three cells +0.406 (t 1.79; 76 clusters); excluding the ten largest clusters +0.404 (t 2.72); leave-one-year-out +0.362 to +0.515 (t 1.87 to 3.09). The 137 tau = 4 cells sit in 137 clusters of 137 firms over 2014-2023. The two-year window (coincidence at firm clock four or five) gives an implied joint effect of +0.622 (SE 0.102, t = 6.13; 1,105 clusters), with a leave-one-year-out range of +0.368 to +0.815, +0.621 (t = 6.05) after dropping the ten largest clusters, and a restricted wild-cluster bootstrap $p < 0.001$ at 9,999 draws.

Cluster-level entropy balancing. The 6,781 non-joint pair-firm clusters are reweighted to the 970 joint clusters on cluster means of size, return on assets, growth, cash, leverage, state ownership, listing age, partner gender mix, education and pay gap, plus cells per cluster and years covered. Largest absolute standardized difference 0.613 before (listing age), 0.00000 after; dual objective -0.540; maximum gradient 7e-10; row weights 0.04 to 15.34 with the control group's total cell mass (16,754 cells) preserved. Weighted Table 8 interaction: D2 +0.140 (t 1.63), D2 x joint +0.608 (t 4.13), implied joint +0.748 (SE 0.120, t 6.23), against +0.057 (t 1.02), +0.596 (t 4.63) and +0.653 (SE 0.115, t 5.66) unweighted; joint-subsample fall unchanged at +1.201 (SE 0.289, t 4.15); non-joint fall -0.149 (t -0.72) against -0.161 (t -1.09). Adding the share of cluster cells at clock four or five to the constraints (pre-balance difference +2.13 standard deviations; row weights to 148): D2 -0.180 (t -1.34), D2 x joint +0.837 (t 4.12), implied joint +0.657 (SE 0.150, t 4.38), non-joint fall -0.444 (t -1.50).

8.27 Firm-level symmetric balance and persistence-constrained reweighting (manuscript Section 4.4, Appendix A.13)

Sample reconstruction. The firm-level battery runs on the same 31,274-firm-year primary sample as Table 6, reconstructed from the deposited main panel (file 18) exactly as Table 2 documents: the clock-keyed panel holds 32,208 rows on 4,666 firms; restricting to partner-clock tenures one to five leaves 31,734 rows, and absorbing singletons (firms with a single retained firm-year) leaves 31,274 rows on 4,166 firms with 2,018 joint firm-years. The Table 6 cell assignment (the audit-firm clock reading in the firm's first partner-clock-five year) then reproduces the published

partition exactly - 1,764 joint, 463 below five, 996 above five and 943 firms with no clock-five year inside the window - and the Table 6 specification reproduces the published cells (+1.192, t = 8.85, joint; +0.104, t = 0.52, below; +0.191, t = 1.32, above), so the battery below is validated against published numbers before any new constraint is added.

Raw balance (manuscript Table A.7). Firm-level covariates are means over the firm's observed firm-years. Against the 2,402 non-joint firms, the 1,764 joint firms differ most on listing age (-0.63 standard deviations; 1.50 against 2.03 on the age index), then on the persistence descriptors: years observed 6.58 against 8.18 (-0.51), years covered 6.92 against 8.60 (-0.51), and presence in all eleven years 15.8 against 38.5 percent (-0.53). Size (-0.31), leverage (-0.21), state ownership (-0.20), return on assets (+0.15), top-ten auditor (+0.15) and cash (+0.14) follow; growth, gender mix, education, pay gap and manufacturing all sit below 0.10. Two quantities are definitional rather than balanceable: every joint firm's partner spell reaches clock five inside the window (1.00 against 0.61), and 44.7 percent of a joint firm's observed years sit at partner clock four or five against 29.4 percent.

Entropy balancing. Weights are Hainmueller and Xu (2013) entropy weights on the 2,402 comparison firms, matching the 1,764 joint firms' means; the treated firms keep weight one and the comparison weights are rescaled to sum to 2,402. Three constraint sets: (i) the nine firm means of Section 8.20 (state ownership, size, return on assets, growth, cash, leverage, age index, top-ten auditor, manufacturing); (ii) the same nine plus years observed and years covered; (iii) the same eleven plus the spell-completion indicator, the firm-level analogue of the clock-four-or-five cell share added at the cluster level in Section 8.26. All three sets balance to machine precision (largest absolute standardized difference: set (i), 6.1e-15; set (ii), 1.5e-14; set (iii), 6.1e-15). Set (i) leaves 1,634 effective comparison firms (weights 0.20 to 4.70); set (ii) leaves 1,629 (0.14 to 4.64); set (iii) drives the 943 firms with no clock-five year inside the window to zero weight and leaves 428 effective comparison firms (weights 0 to 21.7), because completion is one by construction for every joint firm.

A separate battery, not to be combined with sets (i)-(iii). Manuscript Table 10 row 9 reweights a different treated group: the 1,203 balanced firms against the non-balanced remainder, balancing on the ten firm means of the ten controls (size, return on assets, growth, cash, leverage, state ownership, listing age, female directors, executive education, pay gap). That battery leaves 928 effective comparison firms from 2,402 and moves the balanced-minus-non-balanced gap from 1.310 to 0.514 (t = 1.40; Section 8.17). Its 928 and the 1,634 / 1,629 / 428 sequence of sets (i)-(iii) answer different questions on different treated groups - balanced firms against joint firms - and are never pooled or sequenced with one another.

Weighted estimates (manuscript Table A.8). The Table 6 specification (firm and year FE, full controls, firm-clustered) is re-estimated with the firm weights applied to firm-years. The joint-cell fall is +1.182 (SE 0.135, t = 8.79) under set (i), +1.182 (SE 0.135, t = 8.78) under set (ii) and +1.177 (SE 0.135, t = 8.74) under set (iii), against +1.192 (SE 0.135, t = 8.85) unweighted. The joint-minus-below-five gap moves from +1.088 (t = 4.54) unweighted to +0.972 (t = 3.74), +0.993 (t = 3.70) and +0.673 (t = 2.21); the movement in the last row comes from the below-five cell rising to +0.504 (t = 1.84) rather than from the joint cell falling. Set (i) reproduces the Appendix A.8 estimate (+1.182), which validates the weighting scheme.

Persistence-restricted cells. Restricting the Table 6 specification by panel persistence: among the 1,203 firms observed in all eleven years (279 in the joint cell; 13,122 firm-years) the joint-cell fall is +0.044 (SE 0.235, t = 0.19; MDE 0.66 at 80 percent power), the below-five fall -0.407 (t = -1.64) and the joint-minus-below gap +0.451 (t = 1.32); among the remaining 2,963 firms (1,485 in the joint cell; 18,152 firm-years) the joint-cell fall is +1.469 (SE 0.160, t = 9.16; MDE 0.45), the below-five fall +0.816 (t = 2.51) and the gap +0.652 (t = 1.80). The 279 and 1,485 joint-cell firms sum to the 1,764 joint firms. The pair-level counterpart is Section 8.17's balanced-panel joint subsample (271 clusters, -0.096, t = -0.23; MDE 1.17).

Reading. The joint-minus-other differential survives every firm-level reweighting we can construct, including the constraint that holds spell completion fixed; the cell-level fall does not survive persistence and is identified off firms whose spells finish inside the window. The composition concern is therefore narrowed at the firm level, as it was at the cluster level in Section 8.26, but not removed, and the manuscript's framing says so.

9. Archive notes for replicators (deposited package: JAR_SourceData_CSVs)

(i) The audit-panel file stores stock codes as zero-padded strings while the merged and estimation files store integers; the deposited package is type-consistent (all files cast to zero-padded six-character strings). (ii) The archived second-convention E_W copy shows exact equality with E_H in all 10,352 overlapping rows; the raw cross-validation extract (in which 32.7 percent of firm-years differ by rounding or revision noise) is deposited separately. (iii) PairID is the two-name signature as recorded (15,719 distinct strings); sorting the two names within each pair would merge 1,914 order-variant pairs into 13,805 - the deposited package uses the as-recorded convention throughout. (iv) The 15,011 partner count is over the full 1997-2025 audit panel; the clock sample contains 9,628 distinct names. (v) The event-study materials are two files: 12_事件研究样本.csv carries the event-time assignments (et_exact5, et_allchg) under the first-event rule documented in Section 8.3, and 13_修订版表格_事件研究.csv carries the full -5 to +5 coefficient cells quoted in manuscript Section 4.11 and Table 13. (vi) The penalty extraction panel covers 2013-2020 only. (vii) The 2021-2023 rows of the merged panel's environmental-penalty flag and disclosure-hit count are respectively all-zero and all-missing, reflecting the end of the annual-report extraction and not an observed absence of penalties. (viii) The announcement-based penalty indicator in file 16 covers 2013-2023 and is built from the public announcement index alone; it is a disclosure-based measure, not an extension of the licensed register, and the two are not pooled.

9.1 File-name map (Chinese file names to English)

The deposited package JAR_SourceData_CSVs uses bilingual file names for files 00-17 (the mapping is one-to-one; column names inside every file are English); files 18-23 are sheets of this archive workbook and carry the same numbering. Row and column counts and the manuscript table each file reproduces are in the manifest below; files 10-17 are script-generated result tables whose row counts are recorded in the deposit README when the package is assembled, and variable lists are in file 01_变量字典.csv and workbook sheet 01_变量字典. A README listing every file with its SHA-256 checksum accompanies the package at deposit.

File	Contents	Rows x cols	Reproduces
00_源文件清单.csv	Source-file manifest	20 x 7	This manifest
01_变量字典.csv	Variable dictionary (variable, file, meaning, source)	4 cols	All tables; variable list
02_估计样本_32208.csv	Estimation sample, clock-keyed firm-year panel	32,208 x 23	Tables 3-8, 10-13
03_ESG 审计合并面板.csv	Merged ESG-audit panel (supplement layer)	33,923 x 34	Supplement S1; Section 8.24
04_审计签字面板.csv	Audit signing panel 1997-2025 (names, tenure, rotation flags)	88,851 x 18	Table 2 funnel; clocks
05_环保处罚_企业年.csv	Environmental penalties, firm-year (register)	25,749 x 3	Section 4.7; Table 8.4
06_文本环保度量.csv	Textual environmental measures	969 x 16	Section 4.8
07_年报审计匹配.csv	Annual-report audit match detail	1,119 x 17	Section 4.8
08_华证 ESG 交叉验证.csv	Huazheng ESG cross-validation (E vs E_W)	10,352 x 6	Section 8.24
09_配对样本_19376.csv	Pair-by-firm estimation sample	19,376 x 20	Tables 8-9
10_联合推断_WCR 结果.csv	Wild-cluster bootstrap p-values	at deposit	Table 8 inference
11_稳健性估计.csv	Robustness coefficients	at deposit	Table 12; A.3-A.6
12_事件研究样本.csv	Event-study sample, event-time assignments	at deposit	Section 4.11
13_修订版表格_事件研究.csv	Event-study coefficient cells, -5 to +5	at deposit	Table 13
14_组合分析.csv	Composition analysis	at deposit	Table 10
15_扩展检验_三项.csv	Broad flag, turnover shocks, spell matching	at deposit	Sections 4.7, 4.12, 4.13
16_环境处罚公告_企业年.csv	Announcement penalty indicator 2013-2023	at deposit	Section 4.7; Table 8.25

File	Contents	Rows x cols	Reproduces
17_环境处罚公告_估计.csv	Announcement-measure estimates	at deposit	Section 4.7
18_主面板	Master firm-year panel (workbook sheet)	32,208 x 58	All tables rebuild from here
19_Table8 联合分解	Table 8 joint decomposition with SE, t, WCR p, MDE	7 x 10	Table 8; Table 2A
20_平衡面板分解	Balanced-panel composition decomposition	3 x 9	Table 10; Section 4.4
21_企业层平衡检验	Firm-level balance: A.7-A.8 constraint sets	26 x 8	Tables A.7-A.8; A.13
22_正文表格数值	All 24 manuscript tables, cell-level long form	838 x 8	Every table
23_图件作图数据	Figure 1 cell means and counts; Figure 2 levels	44 x 7	Figures 1-2